

# YUNHAI XIANG

July 27, 2026

## PERSONAL DETAILS

|          |                                                                             |
|----------|-----------------------------------------------------------------------------|
| Name     | Yunhai Xiang (in Chinese: 项云海, pronounced “yee-o-en-hi si-ang”)             |
| Pronouns | He/Him/His/Himself                                                          |
| Email    | yxiang72@Western.ca (for encrypted emails: <a href="#">PGP Public Key</a> ) |
| Website  | <a href="#">yunhaimath.com</a>                                              |
| GitHub   | <a href="#">yunhaixiang</a>                                                 |
| Roles    | Researcher, Data Scientist                                                  |

## EDUCATION

|                                                                                                                                                        |                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Western University</b><br><i>Ph.D. in Mathematics (in progress)</i><br>Advisor: Chris Hall<br>Specialization: Number Theory and Arithmetic Geometry | 2025 — Now<br>London, ON, Canada    |
| <b>Western University</b><br><i>M.Sc. in Mathematics</i><br>Advisor: Chris Hall                                                                        | 2024<br>London, ON, Canada          |
| <b>University of Waterloo</b><br><i>B.Math. in Pure Mathematics, Honors, with Distinction</i>                                                          | 2018 — 2023<br>Waterloo, ON, Canada |

## INDUSTRY EXPERIENCE

|                                                                                                                                                                                                                                         |                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>ByteDance</b><br><i>Co-op Internship</i>                                                                                                                                                                                             | 2019<br>Beijing, China |
| <ul style="list-style-type: none"><li>Supported data-informed product development for an education technology product</li><li>Collected and reviewed educational data to support product analysis and application improvement</li></ul> |                        |

## SKILLS

Colored by level of mastery: *Expert*, *Advanced*, *Proficient*, *Familiar*, *Beginner*

|             |                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages   | <a href="#">Mandarin</a> , <a href="#">English</a> , French, Japanese                                                                                                                                   |
| Programming | <a href="#">Python</a> , <a href="#">Lean</a> , <a href="#">C/C++</a> , C#, Rust, Haskell, JavaScript/TypeScript                                                                                        |
| Computing   | <a href="#">SageMath</a> , <a href="#">NumPy</a> , <a href="#">SciPy</a> , <a href="#">Pandas</a> , <a href="#">PyTorch</a> , <a href="#">XGBoost</a> , <a href="#">Sclearn</a> , <a href="#">Slurm</a> |
| DSLs        | <a href="#">LaTeX</a> , <a href="#">SQL</a> , <a href="#">HTML</a> , <a href="#">CSS</a> , GDScript                                                                                                     |
| Softwares   | <a href="#">Cinema 4D</a> , Blender, Unity, Godot                                                                                                                                                       |
| Hobbies     | <a href="#">Magic Tricks</a> , Game Development                                                                                                                                                         |

## PUBLICATIONS

*QEDBench: Quantifying the Alignment Gap in Automated Evaluation of University-Level Mathematical Proofs*  
Joint with Santiago Gonzalez, Alireza Amiri Bavandpour, Peter Ye, Edward Zhang, et al.  
INTERNATIONAL CONFERENCE ON MACHINE LEARNING (ICML) 2026

## AWARDS

## Mathematics Undergraduate Research Award

2022

- Highly competitive undergrad research award of value \$10000 at University of Waterloo. Worked on research project in summer related to Weyl groups, supervised by Prof. Matthew Satriano.

## ACADEMIC MEETINGS & PRESENTATIONS

---

### Conferences Attendance & Presentations

*Attended academic conferences and given research or expository talks*

- Workshop on Noncommutative Geometry and Graph Algebras, Summer 2026, Western
- FrontierMath Open Problems Workshop, Summer 2026, UToronto
- Homotopy Theory, K-theory, and Topological Data Analysis Conference, Summer 2026, Western
- Using AI to Find Examples Workshop, under thematic program Universal Statistics in Number Theory, Winter 2026, Centre de Recherches Mathématiques, Université de Montréal
- University of Western Ontario Research in Computer Science (WesternRCS), Winter 2026, Western
- *Rigid Analytic Geometry*, Ontario Graduate Mathematics Conference, Summer 2024, UWaterloo
- *Cohomology of Grassmannians*, Canadian Undergraduate Mathematics Conference, Summer 2022, Université Laval

### Seminar Attendance & Presentations

*Attended academic seminars and given research or expository talks*

- *Hyperelliptic Jacobians and Hasse–Witt Matrix*, Algebraic Geometry Student Seminar, Western
- *Trace of Frobenius*, Winter 2026, Graduate Student Seminar, Western
- *Brauer Groups of Fields Part II*, Winter 2026, Algebraic Geometry Student Seminar, Western
- *Brauer Groups of Fields Part I*, Winter 2026, Algebraic Geometry Student Seminar, Western
- *Weyl Group and Root Datum*, Summer 2025, Algebraic Groups Seminar, Western
- *Hecke Operators*, Summer 2025, Algebraic Groups Seminar, Western
- *Parabolic and Borel Subgroups*, Summer 2025, Algebraic Groups Seminar, Western
- *Lie Algebras of Algebraic Groups*, Summer 2025, Algebraic Groups Seminar, Western
- *Admissible Topology*, Summer 2024, Rigid Analytic Geometry Seminar, Western
- *Non-Archimedean Fields*, Summer 2024, Rigid Analytic Geometry Seminar, Western
- *Cones and Affine Toric Varieties*, Summer 2024, Toric Varieties Learning Seminar, Western
- *An Easy Tour of Galois Cohomology*, Winter 2024, Graduate Student Seminar, Western
- *Introduction to Derived Categories*, Fall 2022, Algebraic Geometry Working Seminar, UWaterloo

## TEACHING & MENTORING

---

### Course Instructorship

- Coming soon.

### Teaching Assistantship

*Proctored and graded exams, graded homeworks, tutored at Help Centre, and taught tutorial sessions*

- MATH 3150, *Elementary Number Theory I*, Fall 2024
- MATH 2155, *Mathematical Structures*, Fall 2025
- CALCULUS 1501, *Calculus II for Mathematical and Physical Sciences*, Winter 2025, Winter 2026
- MATH 1229, *Methods of Matrix Algebra*, Summer 2024, Summer 2025
- MATH 1228, *Methods of Finite Mathematics*, Summer 2024, Summer 2025
- CALCULUS 1000, *Calculus I*, Winter 2024, Summer 2024, Winter 2025, Fall 2025, Winter 2026

### **Mathematical Outreach Lectures**

*Gave expository lectures to high school students or undergraduate students on advanced mathematical topics*

- *Aperiodic Tilings*, Summer 2026, Western Summer Math Camp (Grade 11/12)
- *Surreal Numbers and the Game of Go*, Summer 2025, Western Summer Math Camp (Grade 10)
- *Mathematical Magic Tricks: Gilbreath's Principle*, Summer 2024, Western Summer Math Camp (Grade 11)
- *Why is There no Quintic Formula*, Summer 2024, Western Summer Math Camp (Grade 10)
- *Category Theory Demystified*, Winter 2023, Short Attention Span Math Seminar (Undergraduate)

### **Directed Reading Program Mentorship**

*Mentored undergraduate students on reading projects, culminating in a presentation*

- Jude McNorgan Lubos, *Introduction to Quadratic Reciprocity*, Winter 2026

## **SERVICE & VOLUNTEER WORK**

---

### **Seminar Organization**

*Research or learning seminars organized or co-organized*

- Algebraic Geometry Student Seminar, Summer 2026
- Algebraic Geometry Student Seminar, Winter 2026, co-organized with Alex Zwart
- Algebraic Geometry Student Seminar, Fall 2024, co-organized with Alex Zwart
- Toric Varieties Learning Seminar, Summer 2024, co-organized with Nikolas Rieke

## **SOFTWARE PROJECTS & OPEN SOURCE CONTRIBUTIONS**

---

Coming soon.

## **CERTIFICATIONS**

---

Coming soon.